

SIMply

Find your spot. Earn your streak.

A Real-Time Campus Study Spot Finder for SIM Students



The Problem Highlights



No Real-Time Visibility

Students walk floor to floor searching for free study spots



Peak-Hour Guesswork

No way to know how crowded a location is before going there



Isolated Studying

No easy way to find classmates studying the same module nearby



Low Campus Engagement

Little incentive to explore different study areas or campus life

Our Solution

Real-Time Campus Data



Interactive Campus Map

SVG floor plan with color-coded pins (green/yellow/red), drag & pinch-to-zoom



QR Check-In / Check-Out

Scan QR at any location, auto-updates occupancy in real time



Real-Time Occupancy

Status auto-calculated from active sessions vs. total seats

Community & Gamification



Gamification Engine

Points, 5 level tiers, missions, streaks, leaderboard



Study Buddy Finder

Create/join study groups by module and location



Rewards Store & Reviews

Redeem points for rewards; rate locations 1-5 stars

Technical Architecture

Frontend

Next.js 16.2.1 + React 19 + TypeScript + Tailwind CSS v4
+ Framer Motion

Backend

Supabase (PostgreSQL + Auth + Realtime)

Deploy

Vercel + Supabase Cloud

Architecture Highlights

- MVC folder structure — /app, /components, /lib/db, /types
- 12 database tables with strict TypeScript types
- Middleware-enforced auth — 3 tiers (guest / user / admin) at the Edge
- DbResult<T> pattern — standardized error handling across all query modules

Key Decisions

SVG over Google Maps

Indoor floor plans, zero API cost

Supabase over custom BE

Built-in auth, real-time, faster dev

CSS variable theming

Auto dark mode via SGT timezone

Project Highlights



Full-Stack TypeScript

Strict type safety from DB to UI, zero 'any' in data layer



Zero-Admin Occupancy

Status auto-updates via DB triggers from check-in/out sessions



Gesture-Rich Map

SVG floor plan with drag, pinch-to-zoom, tap-to-select



Complete Gamification

Points (atomic RPC), levels, streaks, missions, leaderboard, rewards



Auto Dark Mode

Switches at 7PM/6AM SGT via CSS variables, re-checks every 60s



AI Campus Assistant

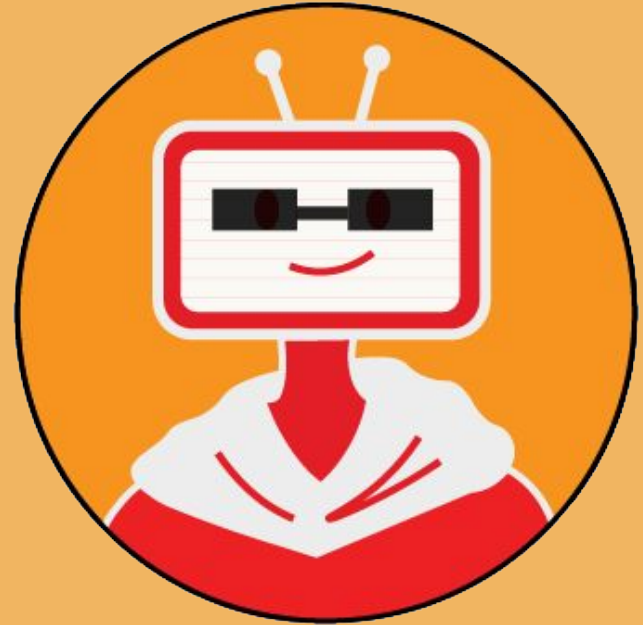
RAG-powered chatbot that reads live database occupancy to recommend study spots.

Thank You

We'd love to hear your questions and feedback.

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<https://itc-simplify.vercel.app/>



“Don’t just try hard, enjoy your life.”
— AkumaV, 2026